

Effective Engine MVP

Variance alpha, neural BSDE hedging, and execution-aware walk-forward retraining

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Project report – 30–45 minutes

C++ options trading engine + Python research pipeline

2026

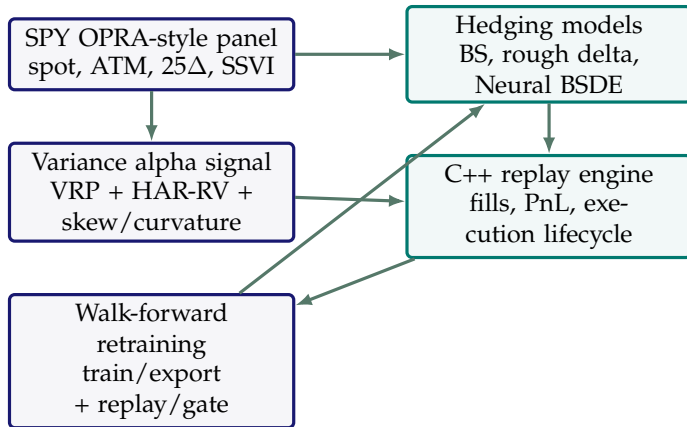
Thirty-Five Minute Map

Part	Main message	Time
Motivation	Options alpha is signal, hedge, execution, and validation.	5 min
State of art	From Black-Scholes to rough vol and deep hedging; deployment remains hard.	7 min
Core idea	Use BSDE learning as a hedge operator, not an alpha destroyer.	10 min
System	C++ replay plus Python walk-forward retraining makes the idea testable.	8 min
Results	Delta-only BSDE preserves most of BS Delta's OOS VRP capture.	7 min
Outlook	Mature pieces, research-grade pieces, and next steps.	3 min

Talk spine

A research prototype for equity-index volatility trading: choose the right hedge objective, replay honestly, and promote only after walk-forward gates.

The Project In One Picture



Main design principle

The project treats alpha, hedge, execution, and model lifecycle as one coupled system.

Market Problem

What Is The Trading Question?

Variance premium signal

$$\text{VRP}_t \approx \sigma_{\text{impl},t}^2 - \sigma_{\text{realized},t}^2.$$

- If implied variance is rich, short front variance can earn carry.
- If realized volatility dominates, long gamma can profit from rebalancing.
- The signal is only useful if hedging and transaction costs do not erase it.

Project implementation

- SPY intraday option-chain replay.
- Model-free VIX-style variance proxy plus ATM fallback.
- SSVI smile features from ATM and 25 Δ quotes.
- Greek attribution and daily stability reports.

State Of Art: Volatility Trading Stack

Layer	Classical / current tools	Why it matters here
Pricing	Black–Scholes, local vol, Heston, stochastic vol [BS73, Hes93, Dup94]	Gives deltas, vegas, and sanity checks.
Surface	SVI/SSVI, skew, curvature, variance swaps [Gat06, Ber15]	Alpha lives in surface dislocations, not only ATM IV.
Dynamics	Rough volatility and Markovian lifts [GJR18]	Short-dated skew has rough-style scaling.
Learning	Deep BSDE and deep hedging [HJE18, BGTW19]	High-dimensional hedging without finite-difference grids.
Deployment	Walk-forward validation, execution replay, risk gates	Research signal must survive data drift and fills.

Data And Signal Construction

Available panel

- SPY spot recovered from option-chain panel.
- ATM call/put quotes with bid/ask.
- 25Δ call/put quotes for skew and butterfly.
- SSVI parameters (ρ, ϕ) and realized vol features.

Composite alpha

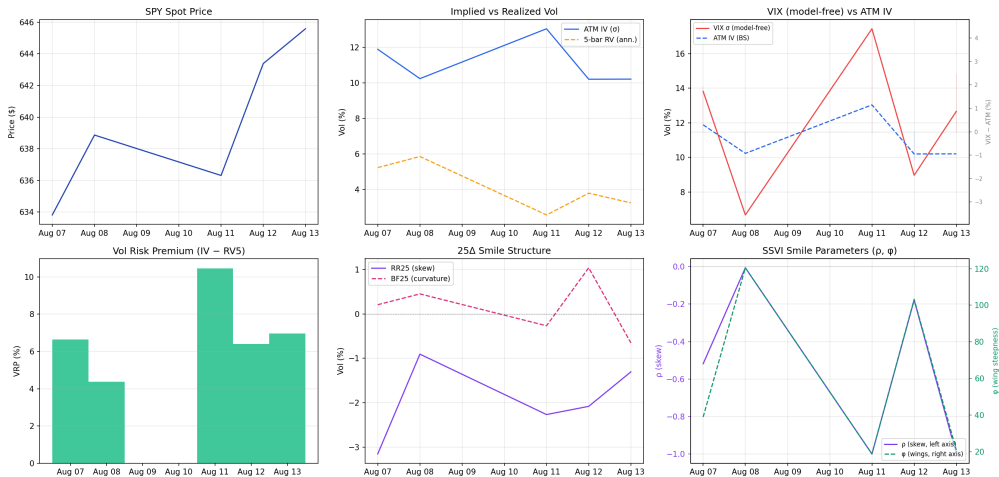
Signal	Weight
VRP	50%
HAR-RV	30%
SkewCurvature	20%

Important choice

Use wings and surface shape; ATM variance alone misses where index-option premium often hides.

Market Context In The Current Panel

SPY Market Context — 5 Trading Days (2025-08-07 → 2025-08-13)



The dashboard tracks spot, ATM IV, realized vol, model-free VIX variance, 25 Δ smile structure, and SSVI dynamics.

Hedging Models

PnL Anatomy: Alpha Versus Hedge

Delta-hedged option PnL, schematically

$$d\Pi_t \approx \underbrace{\Delta_t dS_t}_{\text{spot}} + \underbrace{\frac{1}{2}\Gamma_t d\langle S \rangle_t}_{\text{gamma / realized variance}} + \underbrace{\nu_t d\sigma_t}_{\text{vega / surface move}} + \underbrace{\Theta_t dt}_{\text{decay}} - \text{costs}.$$

- The hedge should remove unwanted spot exposure.
- It should **not automatically remove** the gamma/vega residual that carries the volatility premium.
- This is why the hedging objective is a strategic choice, not only a modeling detail.

Four Hedgers Tested

Hedger	Delta computation
BSDelta	$N(d_1)$ at market ATM IV and historical time-to-expiry.
RoughVolDelta	BS delta plus Bergomi–Guyon smile-slope correction.
NeuralBSDE-Full	ONNX network trained to replicate the full discounted pay-off.
NeuralBSDE-Delta	ONNX network trained to replicate only BS delta hedge PnL.

Hypothesis

A neural hedge can be useful only if its training target matches the trading objective.

Rough Volatility: Why It Enters

State variable

$$X_t = (\tau, \log(S/K), V_t, U_1, U_2, U_3, U_4).$$

- Rough-vol models capture short-time skew scaling.
- Four OU factors approximate the rough kernel in Markovian form.
- The network input is 7D, already beyond comfortable finite-difference grids.

Rough delta correction

$$\Delta_{\text{rough}} = \Delta_{\text{BS}} + \text{Vega} \cdot \frac{\partial \sigma_K}{\partial S}.$$

- More complete hedge.
- But it can consume carry in a positive-VRP regime.

Deep BSDE Hedging: The Modeling Bridge

Zero-driver BSDE recursion

$$Y_{t+\Delta t} = Y_t + Z_t \cdot \Delta W_t, \quad Y_T = \Phi(X_T).$$

- By Feynman–Kac, $Y_t = u(t, X_t)$ and Z_t is the Brownian hedge exposure.
- A shared-weight MLP maps each normalized state to (Y, Z) .
- Export to ONNX makes the hedge available inside the C++ replay loop.

Training mode	Economic meaning
1rh	Full payoff replication; removes nearly all residual alpha.
1rh_delta	Delta-only target; leaves gamma + vega as VRP exposure.

The Key Discovery: Full Replication Can Kill Alpha

Full replication target

$$Y_0 + \sum_i Z_i \Delta W_i \approx e^{-rT} (S_T - K)^+.$$

- The model learns to neutralize payoff risk.
- That includes the gamma/vega residual that the variance strategy wants.

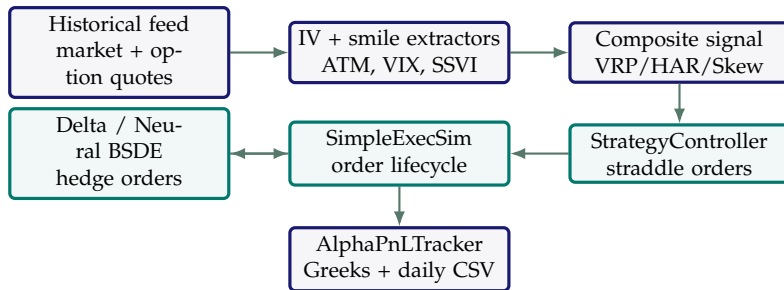
Delta-only target

$$\Phi_\Delta = \sum_i N(d_{1,i}) \sigma_i S_i \Delta W_i.$$

- The model learns the hedge component.
- The residual $\Gamma + \nu$ remains available as alpha.

System Architecture

C++ Event-Driven Replay Architecture



The C++ side is deliberately close to production semantics: events, fills, provenance, risk controls, and replayable PnL.

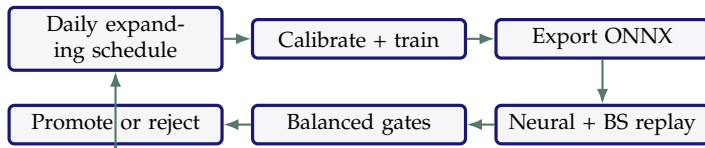
Execution Layer v2: Why It Matters

Area	Before	Now
Hedge fills	Hedgers could self-fill and update positions.	Hedgers submit orders; execution publishes fills.
Provenance	Alpha and hedge fills could blur.	Orders/fills carry producer and order id.
Lifecycle	One-shot fill assumption.	Accepted, rejected, partial, filled, canceled reports.
Risk gates	Mostly logging.	Block, cancel, and reduce-only behavior in router.
Accounting	Position updates could precede execution.	Positions update only from execution fills.

Research implication

Once execution is explicit, turnover and transaction costs become model-selection criteria.

Walk-Forward Retraining Is Now First-Class



Run layout

`demo/runs/walk_forward/<run_id>/windows/<deploy_date>/` stores artifacts, replay CSVs, logs, window manifests, and summary reports.

Promotion Gates

Gate	Checks
Causality	Train only on rows before deploy date; record train/deploy row counts.
Artifact	Checkpoint, ONNX, normalization, Y_0 file, schema, hashes.
ONNX parity	PyTorch versus ONNX output agreement.
Delta sanity	ATM model delta finite and close to BS delta.
Replay health	Neural and BS replays complete and write non-empty daily CSV.
Baseline guard	Neural hedge must not catastrophically underperform BS Delta.

Default behavior

No live artifact is overwritten unless all windows pass and `-promote-if-pass` is explicit.

Results

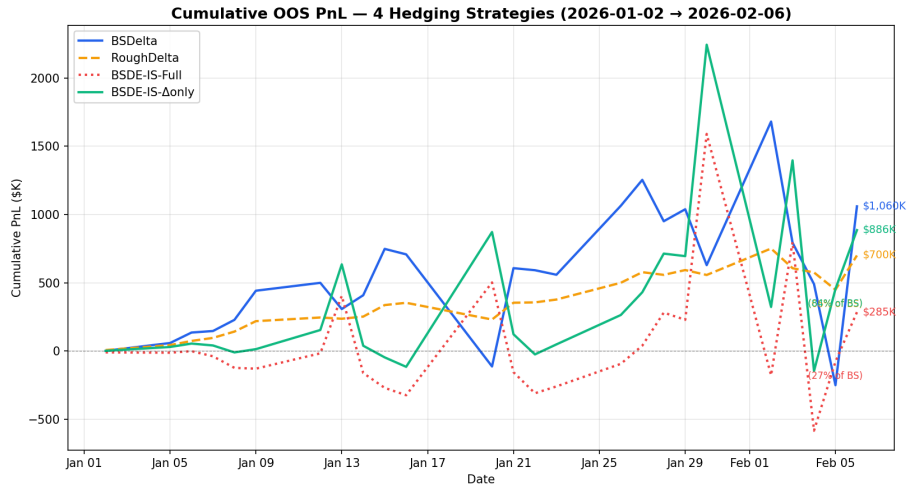
OOS Four-Strategy Result

OOS metric	BSDelta	RoughVol	BSDE-Full	BSDE-Delta
Hedge PnL	1,059,605	698,612	290,817	892,723
Txn cost	1,801	643	8,201	9,005
Total PnL	1,060,036	700,201	284,847	885,950
vs BSDelta	100.0%	66.0%	26.9%	83.6%

Interpretation

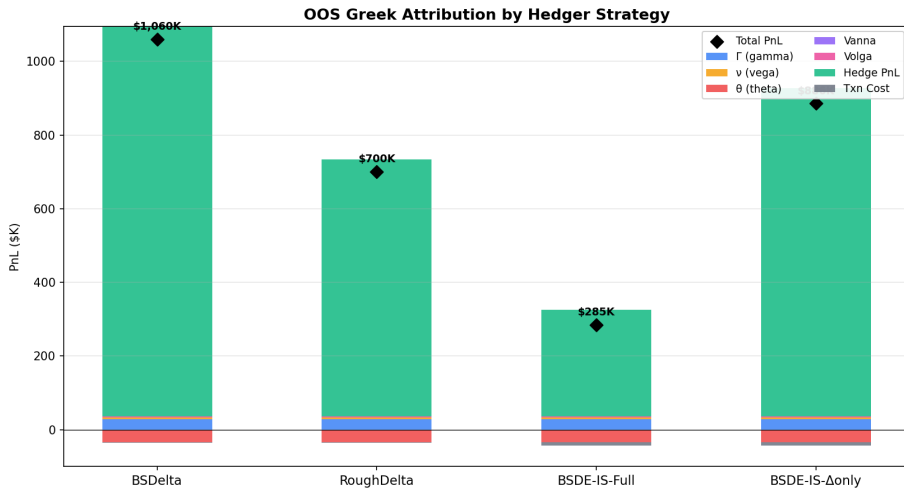
Full replication is theoretically clean but economically wrong for this alpha; delta-only BSDE keeps most of the variance-premium exposure.

Cumulative OOS PnL



The delta-only BSDE tracks BSDelta much more closely than the full-replication BSDE.

Greek Attribution Explains The Difference



Gamma and vega are the intended alpha sources; a hedge objective that erases them also erases the strategy.

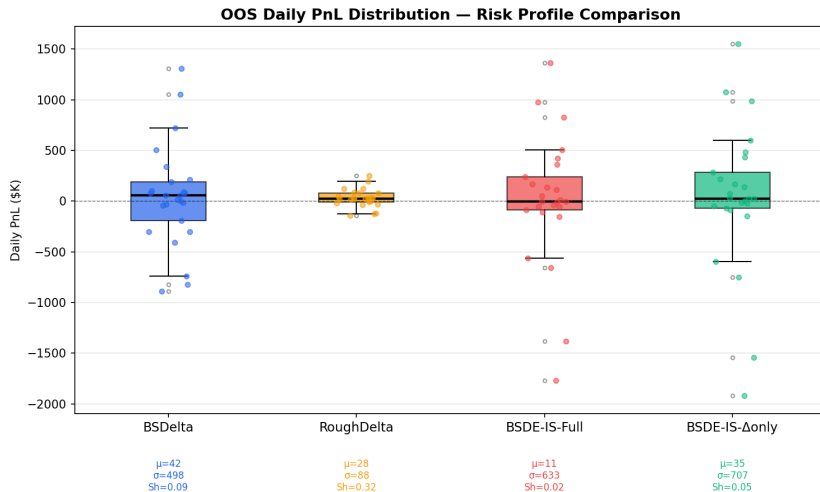
Five-Day Walk-Forward Demo Result

Metric	Value
Total PnL	\$3.415M
Mean daily PnL	\$682,958 $\pm$ \$214,586
Daily Sharpe	3.18
P5 / P50 / P95	\$311K / \$690K / \$929K
Turnover	138.8 fills/day
Model delta vs BS	0.537 vs 0.504

Why walk-forward mattered

The untrained synthetic model was out-of-distribution on SPY; SPY calibration fixed the scale of V_t , strike, maturity, and skew.

Daily PnL Stability



The model comparison is not only final PnL; daily dispersion, transaction costs, and residual risk matter.

Current Status And Outlook

What Is Mature Now?

Solid prototype pieces

- Event-driven C++ replay path.
- Multi-strategy hedge comparison.
- Greek attribution and daily CSV output.
- ONNX inference inside replay.
- Walk-forward run manifests and gates.

Still research-grade

- Small public/local panel in the repo.
- No live broker connectivity.
- Simplified execution simulator.
- BSDE objective still needs broader regime validation.
- Risk sizing and portfolio construction remain MVP-level.

Main Limitations

- **Data scale:** the research claim needs more dates, more regimes, and stricter OOS splits.
- **Cost model:** current fills are replay-realistic but not yet venue/microstructure-realistic.
- **Model objective:** delta-only works better here, but it should be stress-tested against alternative hedge losses and residual-risk metrics.
- **Rough state inference:** U-factors are retained but zeroed at inference to avoid learned confounds.
- **Promotion policy:** gates exist, but production would need monitoring, rollback, and capital/risk budgets.

Roadmap

Next step	Goal
More data regimes	Test positive VRP, negative VRP, stress, quiet, high-rate, low-rate windows.
Better replay costs	Add latency, queue position, venue fee model, and spread impact.
Residual-risk gate	Evaluate hedge residual variance, not only total PnL.
Objective sweep	Compare full payoff, delta-only, residual variance, and transaction-cost-aware losses.
Portfolio layer	Move from one straddle engine to risk-budgeted multi-expiry/multi-strike books.

Takeaway

Main message

The project's major idea is that a neural BSDE should be used as a **hedging operator** inside a full trading system, not as an isolated pricing model.

For volatility alpha, the right question is:

Which risks should the model hedge, and which residuals are the alpha?

The current evidence says full replication can destroy the variance premium, while a delta-only BSDE preserves most of the exposure and can be deployed through a gated walk-forward pipeline.

Thank You

Questions & Discussion

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References I

- [Ber15] Lorenzo Bergomi.
Stochastic Volatility Modeling.
CRC Press, 2015.
- [BGTW19] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood.
Deep hedging.
Quantitative Finance, 19(8):1271–1291, 2019.
- [BS73] Fischer Black and Myron Scholes.
The pricing of options and corporate liabilities.
Journal of Political Economy, 81(3):637–654, 1973.
- [Dup94] Bruno Dupire.
Pricing with a smile.
Risk, 7(1):18–20, 1994.
- [Gat06] Jim Gatheral.
The Volatility Surface: A Practitioner's Guide.
Wiley, 2006.
- [GJR18] Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum.
Volatility is rough.
Quantitative Finance, 18(6):933–949, 2018.

References II

- [Hes93] Steven L. Heston.
A closed-form solution for options with stochastic volatility with applications to bond and currency options.
The Review of Financial Studies, 6(2):327–343, 1993.
- [HJE18] Jiequn Han, Arnulf Jentzen, and Weinan E.
Solving high-dimensional partial differential equations using deep learning.
Proceedings of the National Academy of Sciences, 115(34):8505–8510, 2018.